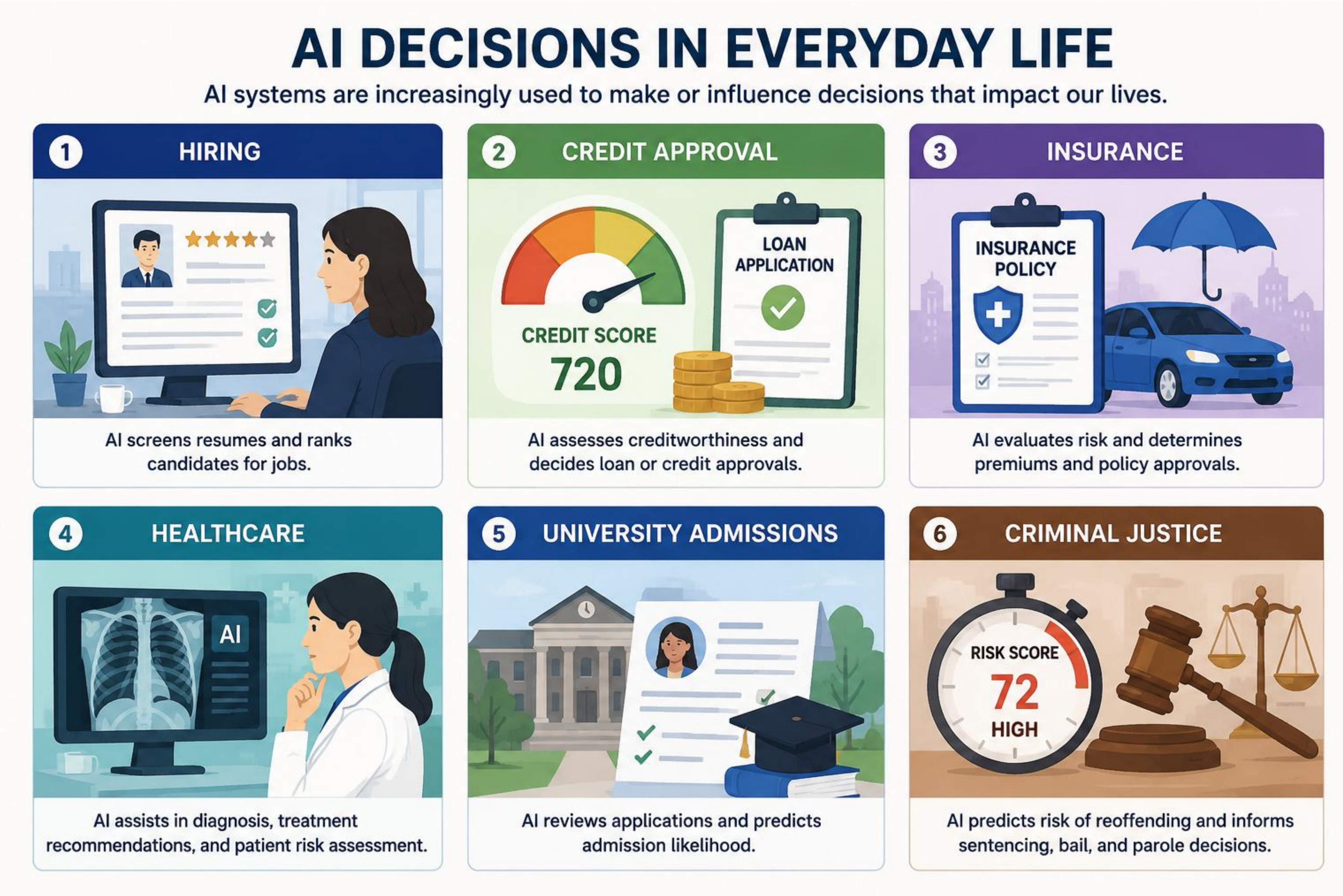
Understanding Bias and Discrimination

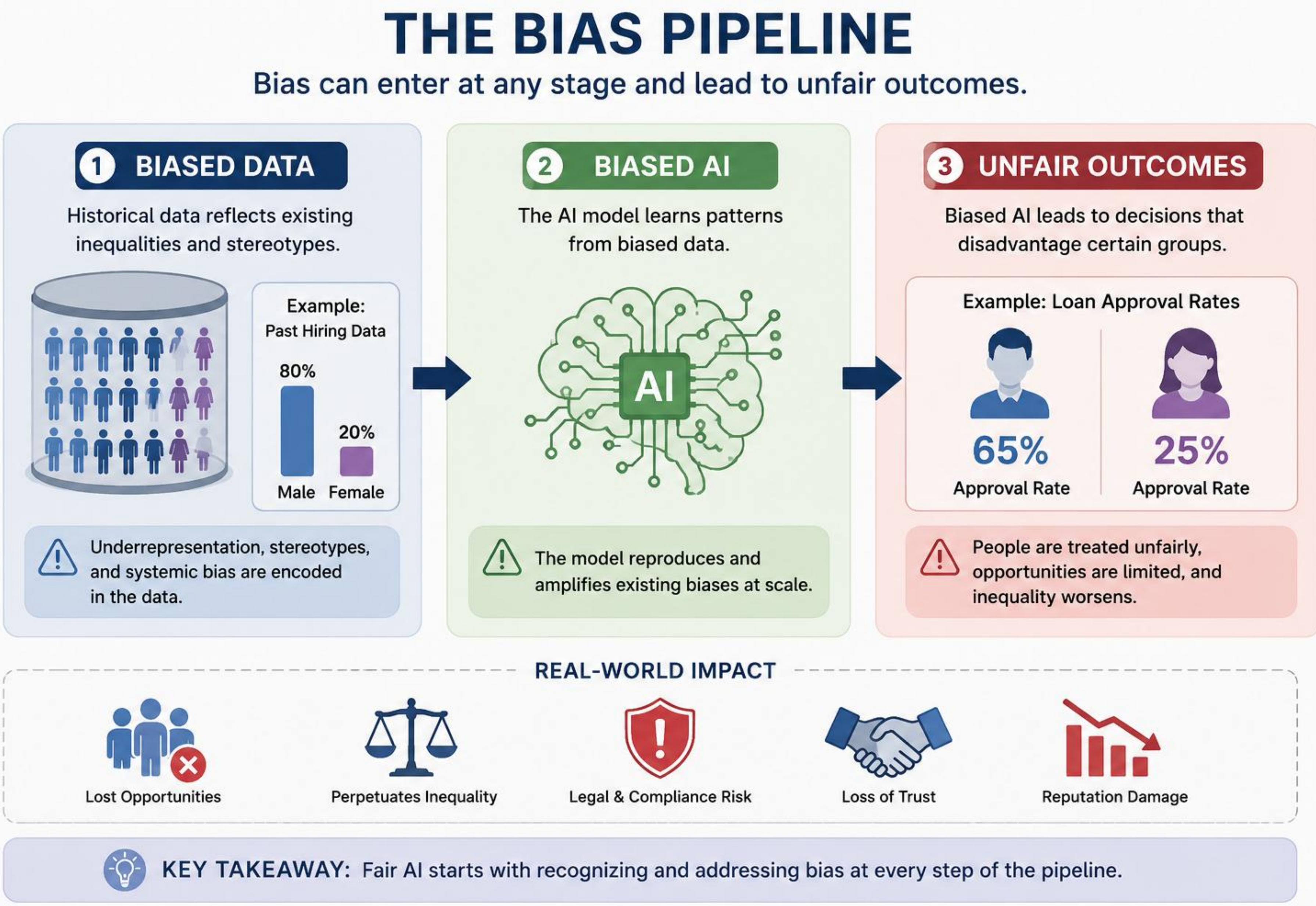
Fairness

AI decisions across: Hiring, Lending, healthcare, admissions and Policing

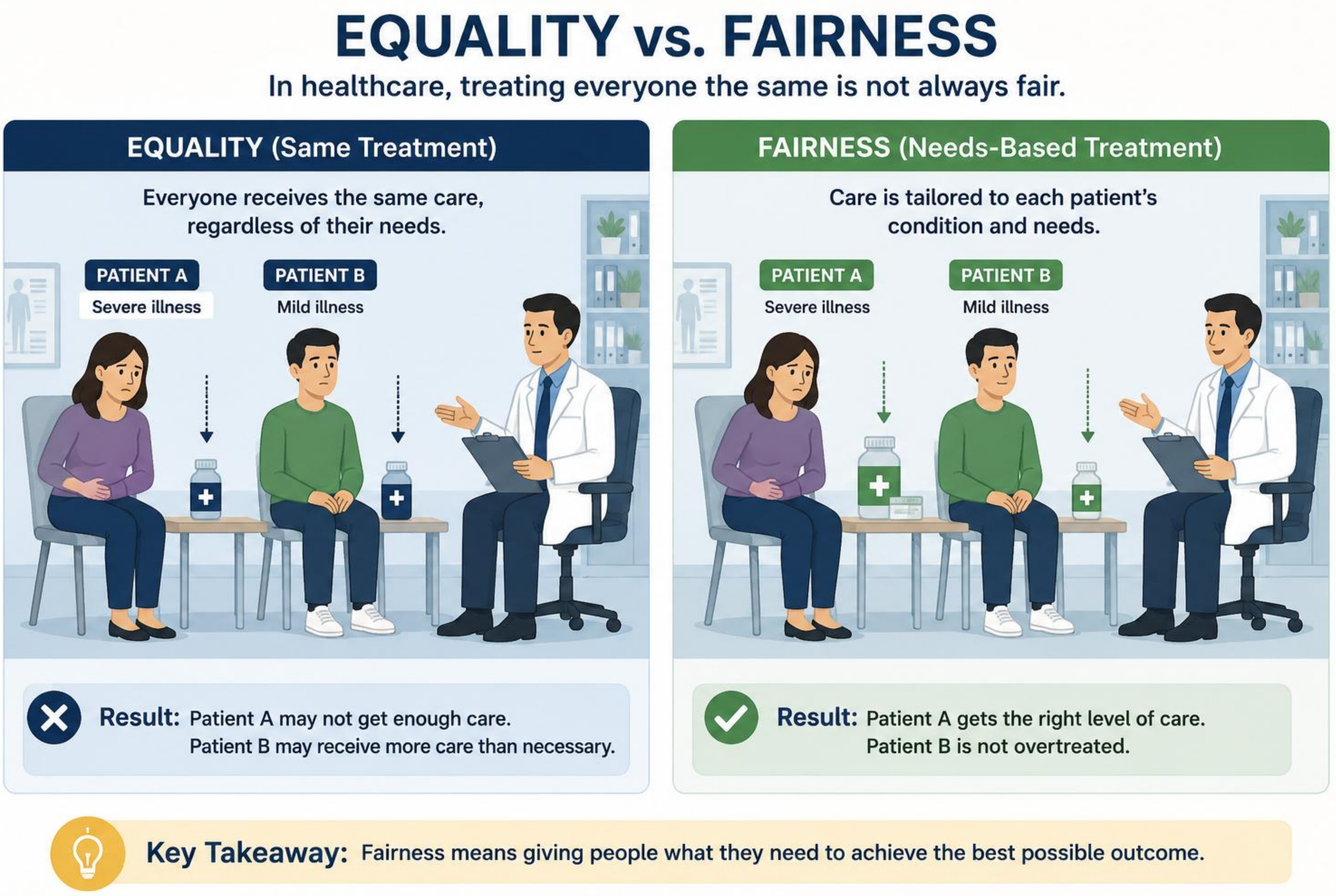
Preventive measures

- Explain algorithmic bias
- Describe fairness metrics
- Identify sources of bias
- Analyze fairness failures
- Discuss mitigation techniques

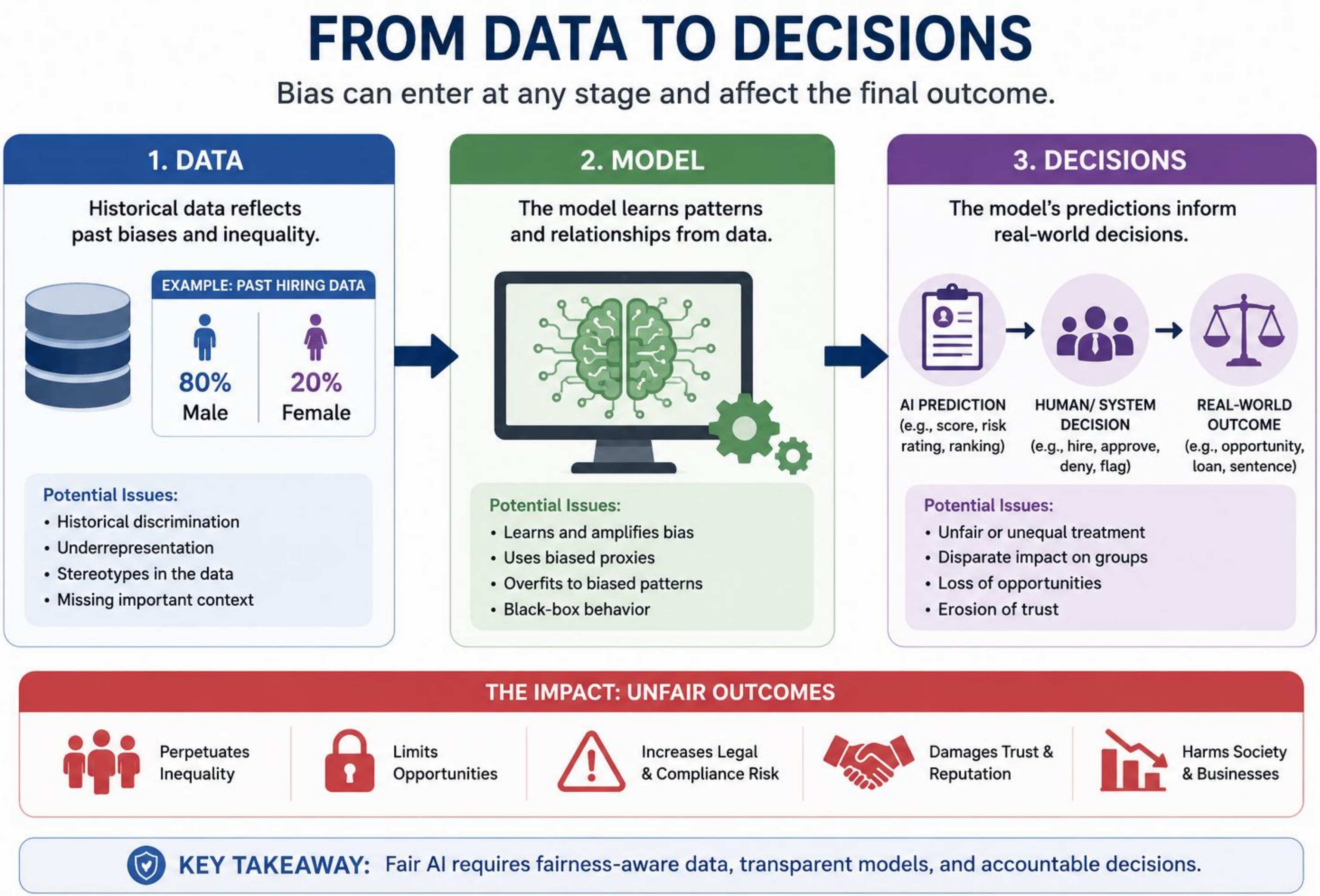

AI DECISIONS IN EVERYDAY LIFE
AI systems are increasingly used to make or influence decisions that impact our lives.
1 HIRING
AI screens resumes and ranks candidates for jobs.
2 CREDIT APPROVAL
CREDIT SCORE
720
LOAN APPLICATION
AI assesses creditworthiness and decides loan or credit approvals.
3 INSURANCE
INSURANCE POLICY
AI evaluates risk and determines premiums and policy approvals.
4 HEALTHCARE
AI
AI assists in diagnosis, treatment recommendations, and patient risk assessment.
5 UNIVERSITY ADMISSIONS
AI reviews applications and predicts admission likelihood.
6 CRIMINAL JUSTICE
RISK SCORE
72
HIGH
AI predicts risk of reoffending and informs sentencing, bail, and parole decisions.


THE BIAS PIPELINE
Bias can enter at any stage and lead to unfair outcomes.
1 BIASED DATA
Historical data reflects existing inequalities and stereotypes.
Example: Past Hiring Data
80%
20%
Male Female
Underrepresentation, stereotypes, and systemic bias are encoded in the data.
2 BIASED AI
The AI model learns patterns from biased data.
AI
The model reproduces and amplifies existing biases at scale.
3 UNFAIR OUTCOMES
Biased AI leads to decisions that disadvantage certain groups.
Example: Loan Approval Rates
65%
Approval Rate
25%
Approval Rate
People are treated unfairly, opportunities are limited, and inequality worsens.
REAL-WORLD IMPACT
Lost Opportunities
Perpetuates Inequality
Legal & Compliance Risk
Loss of Trust
Reputation Damage
KEY TAKEAWAY: Fair AI starts with recognizing and addressing bias at every step of the pipeline.

Fairness ≠ Equality



What Is Algorithmic Bias?



Human Bias vs AI Bias

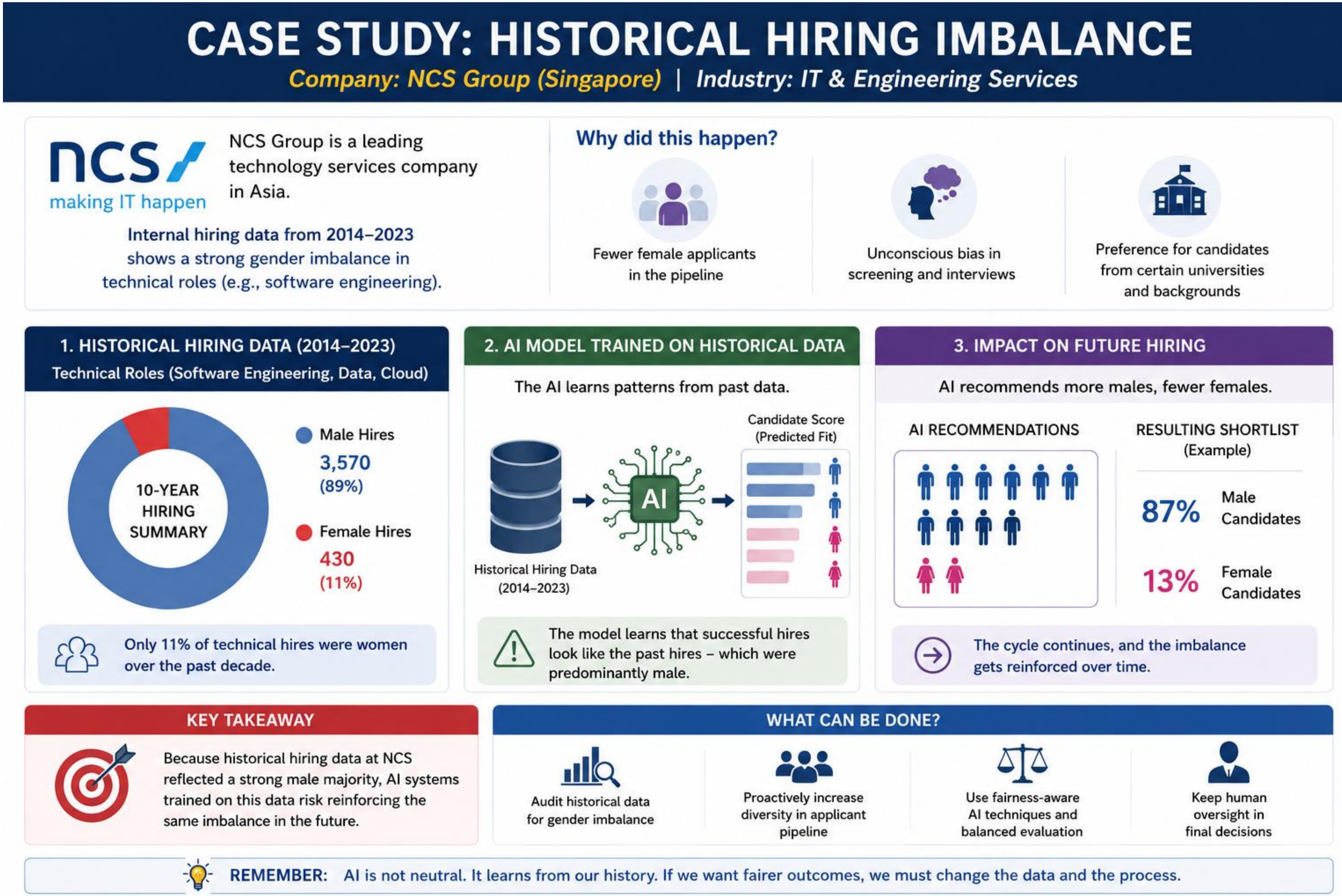
Human Bias	AI Bias
Personal opinions and prejudices	Learns patterns from historical data
Decisions vary from person to person	Decisions are applied consistently
Difficult to measure and audit	Easier to measure and audit
Affects a limited number of decisions	Can affect thousands or millions of decisions
May change over time	Repeats learned biases at scale
Humans can explain their reasoning (sometimes)	AI reasoning may be difficult to understand

Discussion

DISCUSSION: Can an AI system be unbiased if historical data is biased?

Sources of Bias

Bias Type	Simple Definition	Example
Historical Bias	Bias already exists in past decisions or society.	A company historically hired mostly men, so the AI learns to favor male candidates.
Sampling Bias	The training data does not represent the real-world population.	A self-driving car is trained mostly on sunny-weather images and struggles in heavy rain.
Measurement Bias	The data used does not accurately measure what we really want to predict.	Using ZIP code as a proxy for financial stability may introduce socioeconomic bias.
Model Bias	The model or objective function prioritizes one goal and ignores others.	A bank optimizes only for profit and rejects more applicants from certain groups.


CASE STUDY: HISTORICAL HIRING IMBALANCE
Company: NCS Group (Singapore) | Industry: IT & Engineering Services
ncs
making IT happen
NCS Group is a leading technology services company in Asia.
Internal hiring data from 2014–2023 shows a strong gender imbalance in technical roles (e.g., software engineering).
Why did this happen?
Fewer female applicants in the pipeline
Unconscious bias in screening and interviews
Preference for candidates from certain universities and backgrounds
1. HISTORICAL HIRING DATA (2014–2023)
Technical Roles (Software Engineering, Data, Cloud)
10-YEAR HIRING SUMMARY
Male Hires
3,570
(89%)
Female Hires
430
(11%)
Only 11% of technical hires were women over the past decade.
2. AI MODEL TRAINED ON HISTORICAL DATA
The AI learns patterns from past data.
Historical Hiring Data (2014–2023)
AI
Candidate Score (Predicted Fit)
The model learns that successful hires look like the past hires – which were predominantly male.
3. IMPACT ON FUTURE HIRING
AI recommends more males, fewer females.
AI RECOMMENDATIONS
RESULTING SHORTLIST (Example)
87% Male Candidates
13% Female Candidates
The cycle continues, and the imbalance gets reinforced over time.
KEY TAKEAWAY
Because historical hiring data at NCS reflected a strong male majority, AI systems trained on this data risk reinforcing the same imbalance in the future.
WHAT CAN BE DONE?
Audit historical data for gender imbalance
Proactively increase diversity in applicant pipeline
Use fairness-aware AI techniques and balanced evaluation
Keep human oversight in final decisions
REMEMBER: AI is not neutral. It learns from our history. If we want fairer outcomes, we must change the data and the process.

SAMPLING BIAS IN AUTONOMOUS VEHICLES

The training data is not representative of real-world conditions.

TRAINING DATA
Mostly Sunny Days

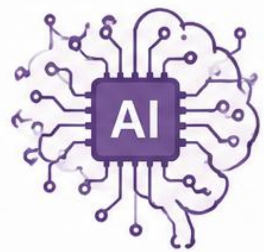
Model trained on 90% sunny day data and only 10% rainy day data

Sunny Days **90%** (90,000 images)

Rainy Days **10%** (10,000 images)

WHAT THE MODEL LEARNS

The model learns patterns that over-represent sunny conditions.



RESULT: Poor generalization to rainy / unseen conditions

REAL-WORLD CONDITIONS
Rainy Days

On rainy days, the vehicle struggles to detect and respond accurately.

CHALLENGES IN RAINY CONDITIONS

- Reduced visibility
- Glare & reflections
- Harder object detection

THE IMPACT

- ✕ Higher risk of accidents in untrained conditions
- ✕ Poor performance in rain, fog, or night
- ✕ Loss of reliability and public trust
- ✕ Regulatory and safety concerns

KEY TAKEAWAY

If the training data does not represent real-world diversity, the AI model will not perform fairly or safely for everyone.

- Collect diverse data across all conditions
- Ensure balanced data distribution
- Continuously test in real-world scenarios
- Monitor and update the model regularly

REMEMBER: A model is only as good as the data it learns from. Biased data leads to biased performance.

CASE STUDY: ZIP CODE AS PROXY FOR CREDITWORTHINESS

When location becomes a hidden predictor of lending decisions

THE SCENARIO

A bank uses an AI model to automate personal loan approvals.

The model includes ZIP code as a feature.

WHY ZIP CODE CAN BE BIASED

- ZIP codes reflect neighborhoods
- Neighborhoods correlate with income, education, and access to resources
- Historical inequalities are embedded in where people live

THE PROBLEM

The model doesn't "see" race or ethnicity, but using ZIP code causes it to indirectly learn patterns tied to socioeconomically disadvantaged communities.

This leads to unfair loan denials.

1. DATA EXAMPLE

Default Rates by ZIP Code (Historical Data)

ZIP Code	Avg. Income	Default Rate
94110	$105,000	2.1%
94122	$98,000	2.3%
94621	$42,000	8.7%
94501	$38,000	9.1%
94550	$36,000	10.2%
94539	$41,000	8.5%

Lower-income neighborhoods have higher historical default rates due to systemic factors, not individual behavior.

2. MODEL LEARNS THE PATTERN

The AI model learns:

"Applicants from certain ZIP codes are more likely to default."

Input Features
- Income
- Employment Length
- Debt-to-Income Ratio
- ZIP Code

→ →

Prediction

Higher risk (from certain ZIP codes)

ZIP code becomes a powerful proxy for credit risk in the model.

3. REAL-WORLD IMPACT

Two applicants with similar financial profiles.

APPLICANT A

Income: $55,000
Debt-to-Income: 25%
Employment: 3 years
ZIP Code: **94122**

APPROVED

Low predicted risk

APPLICANT B

Income: $55,000
Debt-to-Income: 25%
Employment: 3 years
ZIP Code: **94539**

DENIED

High predicted risk

Different outcomes, driven by ZIP code — not by the applicants' actual ability to repay.

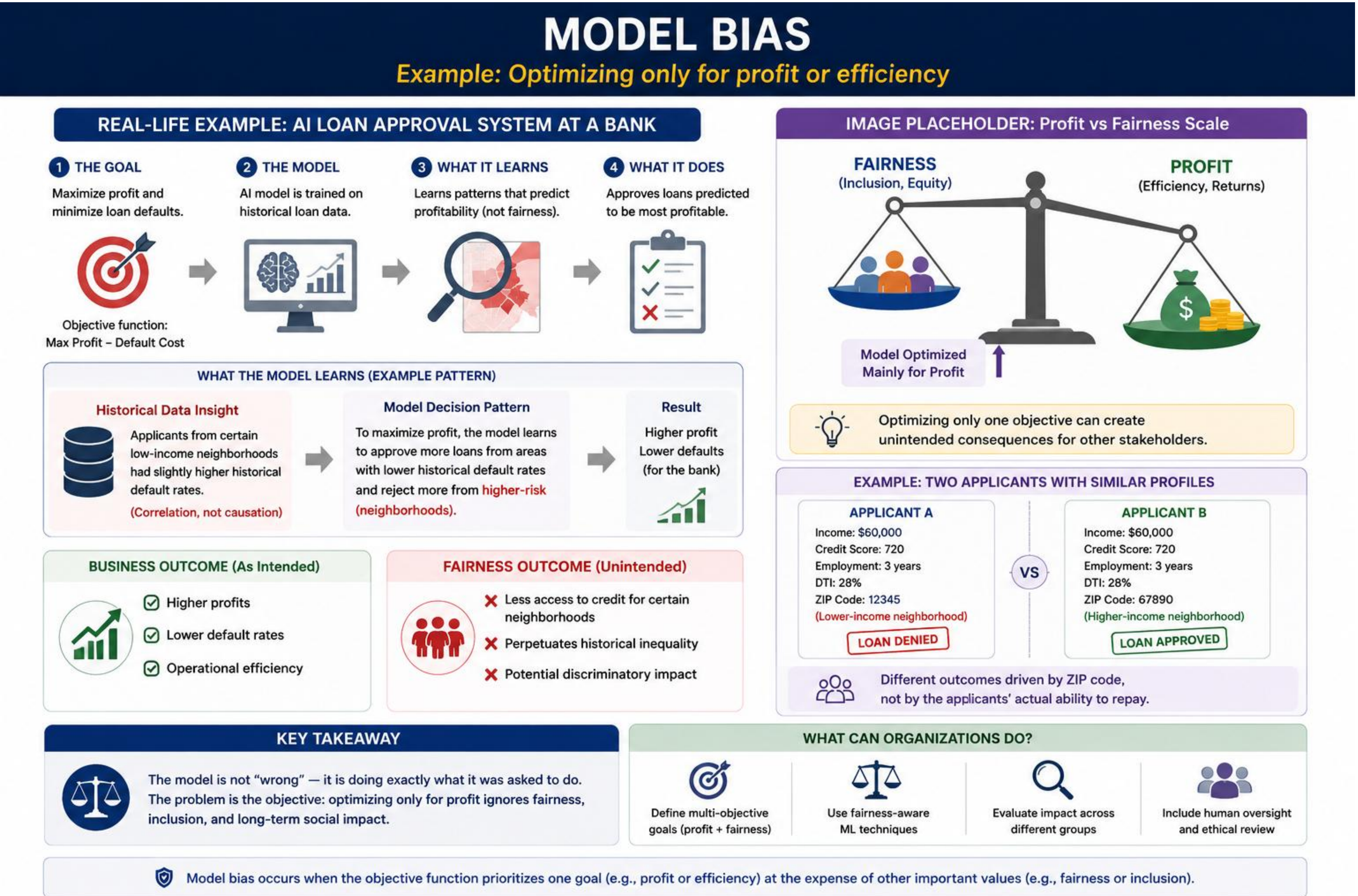
THE BOTTOM LINE

ZIP code is a proxy for race, income, education, and opportunity. Using it in lending models can reinforce historical inequality and lead to discriminatory outcomes, even without explicit intent.

WHAT ORGANIZATIONS CAN DO

- Audit models for proxy variables
- Test for disparate impact across groups and locations
- Use fair lending practices and alternative data
- Ensure human oversight and explainability

KEY TAKEAWAY: Removing protected attributes is not enough. We must also identify and mitigate proxy variables that can embed bias in AI systems.


MODEL BIAS
Example: Optimizing only for profit or efficiency
REAL-LIFE EXAMPLE: AI LOAN APPROVAL SYSTEM AT A BANK
1 THE GOAL
Maximize profit and minimize loan defaults.
Objective function: Max Profit – Default Cost
2 THE MODEL
AI model is trained on historical loan data.
3 WHAT IT LEARNS
Learns patterns that predict profitability (not fairness).
4 WHAT IT DOES
Approves loans predicted to be most profitable.
WHAT THE MODEL LEARNS (EXAMPLE PATTERN)
Historical Data Insight
Applicants from certain low-income neighborhoods had slightly higher historical default rates.
(Correlation, not causation)
Model Decision Pattern
To maximize profit, the model learns to approve more loans from areas with lower historical default rates and reject more from higher-risk (neighborhoods).
Result
Higher profit
Lower defaults
(for the bank)
BUSINESS OUTCOME (As Intended)
Higher profits
Lower default rates
Operational efficiency
FAIRNESS OUTCOME (Unintended)
Less access to credit for certain neighborhoods
Perpetuates historical inequality
Potential discriminatory impact
IMAGE PLACEHOLDER: Profit vs Fairness Scale
FAIRNESS
(Inclusion, Equity)
PROFIT
(Efficiency, Returns)
$
Model Optimized Mainly for Profit
Optimizing only one objective can create unintended consequences for other stakeholders.
EXAMPLE: TWO APPLICANTS WITH SIMILAR PROFILES
APPLICANT A
Income: $60,000
Credit Score: 720
Employment: 3 years
DTI: 28%
ZIP Code: 12345
(Lower-income neighborhood)
LOAN DENIED
VS
APPLICANT B
Income: $60,000
Credit Score: 720
Employment: 3 years
DTI: 28%
ZIP Code: 67890
(Higher-income neighborhood)
LOAN APPROVED
Different outcomes driven by ZIP code, not by the applicants' actual ability to repay.
KEY TAKEAWAY
The model is not “wrong” — it is doing exactly what it was asked to do. The problem is the objective: optimizing only for profit ignores fairness, inclusion, and long-term social impact.
WHAT CAN ORGANIZATIONS DO?
Define multi-objective goals (profit + fairness)
Use fairness-aware ML techniques
Evaluate impact across different groups
Include human oversight and ethical review
Model bias occurs when the objective function prioritizes one goal (e.g., profit or efficiency) at the expense of other important values (e.g., fairness or inclusion).

Scenario

- SmartAid is an NGO that uses AI to allocate disaster relief supplies after a major flood.
- The AI predicts where aid can help the largest number of people.
- Urban areas have larger populations and more available data.
- Rural communities are smaller and have less historical data.

AI Recommendation

- Send 80% of relief supplies to urban areas.
- Send 20% of relief supplies to rural areas.
- The AI believes this maximizes the total number of people helped.

Potential Fairness Concern

- Rural communities may have greater need despite having smaller populations.
- Less data may cause the AI to underestimate their situation.
- Resource allocation becomes driven by efficiency rather than equity.

Transition

Section B: Understanding Fairness Metrics

FAIRNESS PERSPECTIVE DIAGRAM

The same AI decision can be fair to one stakeholder but unfair to another.

AI DECISION CONTEXT

Example: AI system approves loan applications

Goal of the model:
Minimize default risk and maximize profit

THE BANK (LENDER)

"Fairness means minimizing risk and ensuring the bank's sustainability."

What matters most:

- Low default rates
- Financial stability
- Regulatory compliance
- Profitability

Fairness = Responsible risk management

THE APPLICANT

"Fairness means I am treated as an individual, not judged by my ZIP code or background."

What matters most:

- Equal opportunity
- No discrimination
- Individual assessment
- Right to financial inclusion

Fairness = Equal treatment & opportunity

THE COMMUNITY ADVOCATE

"Fairness means the system does not reinforce historical disadvantage or structural inequality."

What matters most:

- Equitable outcomes across groups
- Addressing historical bias
- Reducing disparities
- Long-term community well-being

Fairness = Equity & justice

THE REGULATOR

"Fairness means complying with laws and protecting individuals from unfair or discriminatory outcomes."

What matters most:

- Legal compliance
- Non-discrimination
- Transparency & accountability
- Consumer protection

Fairness = Legal & procedural fairness

THE DATA SCIENTIST

"Fairness means the model is measured properly and performs well across groups."

What matters most:

- Fairness metrics (e.g., parity, equality)
- Robust, unbiased data
- Model accuracy & reliability
- Explainability

Fairness = Measurable & technical fairness

KEY TAKEAWAY: Fairness is not one-size-fits-all. It depends on values, goals, and the impact on each stakeholder.
Good AI systems consider multiple perspectives and balance competing fairness goals.

Demographic Parity

Definition

•Equal acceptance rates across groups

•E.g. Hiring outcomes by gender

Simple Explanation

Demographic parity is one of the simplest fairness metrics. It says that the percentage of people receiving a positive outcome should be similar across different groups. In a hiring example, if 60% of male applicants are selected, then approximately 60% of female applicants should also be selected. The focus is on **equal outcomes**, regardless of the reasons behind the decisions. **Example below: Which table shows parity has been achieved?**

Group	Applicants	Hired	Hiring Rate
Men	100	60	60%
Women	100	60	60%

Group	Applicants	Hired	Hiring Rate
Men	100	70	70%
Women	100	40	40%

Limitations of Demographic Parity

TRADEOFF ILLUSTRATION: THE LIMITATIONS OF DEMOGRAPHIC PARITY

Demographic parity requires equal acceptance rates across groups.

Focus: Equal acceptance rates across groups
Metric: P(Selected | Group A) = P(Selected | Group B)

SCENARIO 1: DEMOGRAPHIC PARITY ACHIEVED
Equal acceptance rates across groups

Group (Gender)	Applicants	Selected (Hired)	Acceptance Rate
Men	100	60	60%
Women	100	60	60%

Demographic parity is satisfied.
Outcomes are identical across groups.

BUT, LOOK CLOSER...
Women in this applicant pool were, on average, more qualified than men.

Potential concern:
Enforcing identical outcomes may lead to hiring less qualified men and overlooking more qualified women.

SCENARIO 2: DEMOGRAPHIC PARITY NOT ACHIEVED
Different acceptance rates across groups

Group (Gender)	Applicants	Selected (Hired)	Acceptance Rate
Men	100	70	70%
Women	100	40	40%

Demographic parity is not satisfied.
Men are hired at a higher rate than women.

BUT, LOOK CLOSER...
Here, men in this applicant pool were, on average, more qualified.

Potential concern:
Lower selection rate for women may reflect true differences in qualifications—or it may reflect bias. Need further analysis.

WHY THIS SHOWS A LIMITATION OF DEMOGRAPHIC PARITY

- It only looks at outcomes, not qualifications or merit.
- It can require admitting less qualified candidates from one group
- It can mask important differences in applicant pools.
- Treating everyone the same (equality) is not always the same as treating everyone fairly (equity).

Other fairness goals may:
- ✓ Consider qualifications (e.g., Equal Opportunity)
- ✓ Balance errors (e.g., Equalized Odds)
- ✓ Ensure individual fairness

PROVOKE A THOUGHT: SHOULD OUTCOMES ALWAYS BE IDENTICAL?

Fairness is about more than equal numbers—it's about what is right for the context.

EQUAL TRUE POSITIVE RATES ACROSS GROUPS

A fairness criterion: give everyone with the positive outcome the same chance of being correctly identified.

THE IDEA

Equal True Positive Rate (TPR) means that the proportion of actual positives that are correctly predicted (caught) is the same across groups.

In simple terms:

If someone truly has the positive outcome, they should have the same chance of the model correctly identifying them, no matter which group they belong to.

Focus: Treating people with the same need similarly, across groups.

CONFUSION MATRIX (for the positive class)

PREDICTED BY MODEL \ ACTUAL (GROUND TRUTH)	Positive (Has risk)	Negative (No risk)
Positive (Predicted risk)	True Positive (TP) Correctly identified as risk	False Positive (FP) Incorrectly flagged as risk
Negative (Predicted no risk)	False Negative (FN) Missed: should have been flagged as risk	True Negative (TN) Correctly identified as no risk

TRUE POSITIVE RATE (TPR) / SENSITIVITY

$$TPR = \frac{TP}{TP + FN}$$

Out of all actual positives (with risk), how many did we correctly identify?

HOW TO READ THIS EXAMPLE

- In Group A (Men): 80 out of 100 people who actually default were correctly identified (TPR = 80%).
- In Group B (Women): 48 out of 60 people who actually default were correctly identified (TPR = 80%).
- Therefore, the model gives equal true positive rates across groups.

WHAT IF TPRs WERE DIFFERENT?

If TPRs differ, one group's actual positives are more likely to be missed (higher FN rate).

Group	TPR	Interpretation
Men	80%	Baseline
Women	60%	More women who will default are being missed.

EXAMPLE: Loan Default Prediction

Goal: Predict whether a loan applicant will default (positive class = Default)

GROUP A: Men

Actual	Actual Default (Positive)	Actual No Default (Negative)	Total
Predicted Default (Model says: Default)	TP = 80	FP = 30	110
Predicted No Default (Model says: No Default)	FN = 20	TN = 190	210
Total	100	220	320

TPR (Men) = TP / (TP + FN) = 80 / (80 + 20) = 0.80 (80%)

GROUP B: Women

Actual	Actual Default (Positive)	Actual No Default (Negative)	Total
Predicted Default (Model says: Default)	TP = 48	FP = 24	72
Predicted No Default (Model says: No Default)	FN = 12	TN = 116	128
Total	60	140	200

TPR (Women) = TP / (TP + FN) = 48 / (48 + 12) = 0.80 (80%)

Equal TPR Achieved: Both groups have a TPR of 80%.
Among people who will actually default, the model is equally good at identifying them in both groups.

WHY IT MATTERS

Equal TPR helps ensure the model is equally accurate at finding people who truly need attention, across groups.

WHEN TO USE

Useful in high-stakes settings where missing a positive case has a serious cost for individuals (e.g., healthcare diagnosis, fraud detection, recidivism risk).

NOT A PERFECT SOLUTION

Equal TPR is one fairness criterion. Other criteria (like equal FPR, equal outcomes) may lead to different trade-offs.

KEY TAKEAWAY

Equal True Positive Rate = Equal chance of catching actual positives across groups.

EQUAL OPPORTUNITY: AN EXAMPLE

Qualified candidates receive equal chances.

1. WHAT IS EQUAL OPPORTUNITY?

Equal Opportunity requires that the model identifies qualified candidates at the same rate across different groups.

Focus on the truly qualified (positive class).
Among people who are actually qualified, everyone should have an equal chance of being correctly selected.

2. HIRING EXAMPLE CONTEXT

Goal: Identify qualified candidates (Good Fit) for the job.
Positive outcome = Candidate is qualified and selected.

3. CONFUSION MATRIX (Positive = Qualified)

ACTUAL (GROUND TRUTH)

PREDICTED BY MODEL	Qualified (Good Fit)	Not Qualified (Not a Good Fit)
Selected (Model says: Hire)	**True Positive (TP)** Qualified candidate correctly selected	**False Positive (FP)** Not qualified, but selected
Not Selected (Model says: Do not hire)	**False Negative (FN)** Qualified candidate missed (not selected)	**True Negative (TN)** Not qualified, correctly not selected

4. HIRING EXAMPLE: EQUAL OPPORTUNITY IN ACTION

Qualified candidates are the "positive" group.

GROUP A: MEN

	Actual Qualified	Actual Not Qualified	Total
Selected (Hire)	TP = 80	FP = 20	100
Not Selected (Do not hire)	FN = 20	TN = 80	100
Total	100	100	200

TPR (Men) = TP / (TP + FN) = 80 / (80 + 20) = 80%

GROUP B: WOMEN

	Actual Qualified	Actual Not Qualified	Total
Selected (Hire)	TP = 64	FP = 16	80
Not Selected (Do not hire)	FN = 16	TN = 64	80
Total	80	80	160

TPR (Women) = TP / (TP + FN) = 64 / (64 + 16) = 80%

5. INTERPRETATION

Equal Opportunity Achieved!
Both groups have the same True Positive Rate (TPR = 80%).

Among all the truly qualified candidates...

- The model correctly selects 80% of men.
- The model correctly selects 80% of women.

Therefore, qualified candidates receive equal chances of being identified and selected.

KEY TAKEAWAY
Equal Opportunity focuses on treating qualified individuals fairly—giving them an equal chance to be recognized.

WHY IT MATTERS
Missing qualified candidates (False Negatives) means losing great talent from one group more than another.

KEEP IN MIND
Equal Opportunity does not ensure equal hiring rates overall. It only ensures equal identification of the truly qualified.

THINK ABOUT IT
Even when overall hiring rates differ, are we giving every qualified person the same chance to succeed?

Scenario

A company uses AI to screen job applicants.

- Men hired: **70%**
- Women hired: **50%**
- Qualified men selected: **80%**
- Qualified women selected: **80%**

Discussion Question

Is this system fair?

- Should fairness mean **equal hiring rates** across groups?
- Should fairness mean **equal chances for qualified candidates**?
- Can both goals always be achieved at the same time?
- Which definition would you choose as a regulator, employer, or applicant?

Factor	Family A	Family B
Household Income	SGD 6,000	SGD 6,100
Family Size	4	4
Citizenship Status	Same	Same
Housing Type	Same	Same

If Family A receives a grant but Family B is rejected solely because of a very small income difference, citizens may perceive the decision as unfair because the two cases are almost identical.

Key Takeaway
Individual Fairness asks whether similar people are treated similarly. If two applicants, borrowers, or citizens are nearly identical, the AI should generally produce similar outcomes unless there is a clear and justifiable reason not to.

GROUP FAIRNESS & GROUP COMPARISON

Fairness is not just about individuals, but also about how outcomes are distributed across groups.

Singapore Example: AI system that predicts "High Risk of Missing School" to trigger early support for students.

1. WHAT IS GROUP FAIRNESS?

Group fairness means the AI system performs similarly well for all groups.

- We care about error rates and selection rates across different student groups.

2. WHAT IS GROUP COMPARISON?

Group comparison means we measure and compare these rates between groups.

- If there are large differences, the system may be unfair to some groups.

3. SINGAPORE CONTEXT & EXAMPLE

The Ministry of Education (MOE) uses an AI model to identify students who may be at risk of missing school so that timely support can be provided.

Student Groups (by Ethnicity)

- Group A: Chinese
- Group B: Malay
- Group C: Indian
- Group D: Others

4. GROUP COMPARISON METRICS

A. SELECTION RATE
(Flagged as "High Risk")

Group	Students	Flagged	Selection Rate
Chinese	10,000	900	9.0%
Malay	5,000	750	15.0%
Indian	4,000	480	12.0%
Others	1,000	90	9.0%

B. TRUE POSITIVE RATE (TPR)
(Among students who actually went on to miss school)

Group	Actual At Risk	Correctly Flagged (TP)	TPR
Chinese	1,000	700	70%
Malay	900	540	60%
Indian	800	480	60%
Others	200	120	60%

5. INTERPRETATION

Selection Rate is much higher for Malay and Indian students.
→ They are flagged more often as "High Risk".

True Positive Rate is lower for Malay and Indian students.
→ Among students who actually miss school, the AI is less effective at identifying them.

This indicates potential **unfairness:** the model may over-flag some groups and under-detect risk in others.

6. WHY IT MATTERS

- Group fairness ensures no student group is systematically disadvantaged by the AI system.
- Ensures public resources (interventions, counselling, mentoring) are allocated fairly.
- Builds trust in AI systems used in public services in Singapore.

Key Takeaway

Group fairness looks at outcomes across groups.

Group comparison helps us measure and detect unfair differences.

Fair AI in Singapore means every student, regardless of race or background, has an equal chance to be correctly identified and supported.

Comparing Fairness Metrics

Fairness Metric	Main Question	Focus	Hiring Example	Strength	Limitation
Demographic Parity	Are outcomes equal across groups?	Equal outcomes	60% of men hired, 60% of women hired	Easy to understand and measure	Ignores qualifications
Equal Opportunity	Are qualified people treated equally?	Equal True Positive Rates	80% of qualified men selected, 80% of qualified women selected	Focuses on deserving candidates	Does not guarantee equal outcomes
Individual Fairness	Are similar people treated similarly?	Similar individuals	Two nearly identical applicants receive similar hiring decisions	Intuitive and human-centric	Difficult to define "similar"
Group Fairness	Are groups treated fairly overall?	Group-level outcomes and errors	Compare hiring rates and error rates across groups	Detects systematic disparities	May hide unfairness at the individual level

- Not all metrics can be satisfied simultaneously

Simple Hiring Example

Suppose:

Group A

•80% qualified applicants

Group B

•50% qualified applicants

Now imagine we want:

Demographic Parity

```
60% hired from Group A
60% hired from Group B
```

To achieve this, we may need to reject some qualified candidates from Group A or hire more less-qualified candidates from Group B.

Why Fairness Metrics Conflict

- Not all metrics can be satisfied simultaneously

Equal Opportunity

Instead, suppose we focus on:

```
80% of qualified applicants selected
for both groups
```

Now hiring rates may become:

```
Group A = 64%
Group B = 40%
```

Equal Opportunity is satisfied.

Demographic Parity is not.

The Key Insight

The same AI system may be:

✅ Fair under one definition

❌ Unfair under another definition

depending on which metric we choose.

Yes, this idea is closely related to the work of:

- Jon Kleinberg
- Sendhil Mullainathan
- Manish Raghavan

and separately

- Chouldechova

around 2016–2017.

Their research showed that when groups have different underlying base rates, it is often mathematically impossible to satisfy multiple fairness criteria simultaneously.

ACCURACY AND FAIRNESS MAY CONFLICT

Improving overall accuracy does not always mean being fair to every group.

Example (Singapore Context): An AI system predicts students who are at risk of missing school so that early support can be provided.

MODEL A: HIGH ACCURACY (OVERALL)

Not Fair Across Groups

Results by Group

Group	Accuracy	True Positive Rate (TPR)	Flagged as High Risk
Chinese	92%	85%	10%
Malay	78%	55%	25%
Indian	76%	50%	23%
Others	90%	82%	12%

Overall Accuracy

87%

Unfair: Malay and Indian students are flagged more often, but the model is less accurate in identifying those who truly need help.

MODEL B: FAIRER (EQUAL OPPORTUNITY)

Lower Overall Accuracy

Results by Group

Group	Accuracy	True Positive Rate (TPR)	Flagged as High Risk
Chinese	80%	70%	18%
Malay	80%	70%	18%
Indian	80%	70%	18%
Others	80%	70%	18%

Overall Accuracy

80%

Fairer: All groups have the same chance of being correctly identified (equal TPR).

Why the Conflict?

- Group base rates are different (e.g., different proportions of students who actually miss school).
- To achieve high accuracy, the model treats groups differently.
- To treat groups fairly, the model must use similar thresholds, which can reduce overall accuracy.

Base Rate (Actual at Risk)

Different across groups

What This Means

There is a trade-off between accuracy and fairness. You can improve one, but it may reduce the other.

We need to decide which goal matters more for the application and the stakeholders.

KEY TAKEAWAY

Accuracy and fairness may conflict.
A perfectly accurate model is not necessarily a fair model.

The right choice depends on the context, values, and impact on each group.

Transition

Section C: Real-World Fairness Failures

CASE STUDY: AMAZON HIRING FAILURE

AI Penalized Resumes Associated with Women

amazon

BACKGROUND

Amazon developed an AI system to automate screening of resumes for technical roles.

The goal was to hire the "best talent" based on historical hiring data.

THE PROBLEM

The AI learned from 10 years of historical data that was dominated by men.

As a result, the model learned to prefer male-associated resumes and penalize those associated with women.

HOW THE BIAS SHOWED UP

Signals Associated with Women	Impact on AI Scoring
Words like "women's", "female", "she", "her"	Lower score ↓
Membership in women's groups (e.g., Women in Engineering)	Lower score ↓
Graduated from women's colleges	Lower score ↓
Resume templates that included the word "executed" more often (found more in male resumes)	Higher score ↑

EXAMPLE: SAME QUALIFICATIONS, DIFFERENT SCORES

RESUME A (MALE-ASSOCIATED)

- BS Computer Science
- 5 years of experience
- Led projects
- Executed solutions
- Strong engineering skills

AI SCORE: 92/100

RESUME B (FEMALE-ASSOCIATED)

- BS Computer Science
- 5 years of experience
- Led projects
- Collaborated on solutions
- Strong engineering skills

AI SCORE: 62/100

THE IMPACT

The system downgraded a **large** number of resumes that included female-associated language or backgrounds.

Women applicants were less likely to be shortlisted, regardless of their actual qualifications.

Internal evaluation showed the **AI penalized** resumes containing words like "women's" or "female" in ~57% of cases.

WHAT AMAZON DID

- Discontinued the AI recruiting tool in 2018.
- Conducted an internal review of the model.
- Strengthened fairness awareness and auditing in AI/ML practices.
- Continues to invest in tools for fairness, transparency and accountability.

KEY TAKEAWAY

AI systems learn from historical data. If that data reflects past biases, the AI will learn and amplify those biases.

Fair AI requires fair data, careful evaluation, and continuous monitoring.

This case is a reminder that AI is not neutral. Without fairness by design, AI can perpetuate and even worsen inequality.

The goal is not just automation, but **fair and inclusive outcomes** for all candidates.

Why Amazon Failed

The Root Cause

•The AI was not explicitly programmed to discriminate.
•It learned bias from historical data.
•Historical bias became model bias.

Impact

•Qualified female candidates were ranked lower.
•The system risked reinforcing existing gender imbalances.
•Amazon eventually abandoned the recruiting tool.

Lessons Learned

•AI learns from historical data, including historical mistakes.
•Removing gender alone may not remove bias.
•Fairness audits are essential before deployment.
•Historical data should never be assumed to be neutral.

CASE STUDY: COMPASS RECIDIVISM JUSTICE FAILURE

COMPAS

When an AI risk assessment tool systemically over-predicted risk for Black defendants.

WHAT IS COMPAS?

- COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is an AI tool used in US courts.
- It predicts the likelihood that a defendant will reoffend (recidivate).
- Scores help judges make decisions about bail, sentencing, and supervision.

THE PROBLEM

- COMPAS was found to be biased against Black defendants.
- It systematically over-predicted the risk of reoffending for Black defendants compared to White defendants.
- This led to unfair court outcomes.

THE IMPACT

- Black defendants were more likely to be classified as "High Risk".
- They were more likely to be denied bail or given harsher sentences.
- Reinforced racial disparities in the criminal justice system.

WHAT THE EVIDENCE SHOWED

FALSE POSITIVE RATE
(Predicted High Risk, But Did Not Reoffend)

Black Defendants 45%

White Defendants 23%

Black defendants were almost twice as likely to be incorrectly labeled High Risk.

TRUE POSITIVE RATE
(Predicted High Risk, And Did Reoffend)

Black Defendants 28%

White Defendants 44%

White defendants who were labeled High Risk were more likely to actually reoffend.

HOW DID THIS HAPPEN?

Historical Bias in Data
The training data reflected past policing and sentencing biases against Black communities.

AI Learned the Bias
The model learned patterns from biased data, not true indicators of future crime.

Biased Outcomes
The biased predictions influenced real-world decisions, perpetuating racial inequality.

REAL WORLD CONSEQUENCES

Higher bail amounts
Black defendants were more likely to be denied bail.

Longer sentences
Over-prediction of risk led to harsher sentencing.

Widening disparities
It deepened racial inequities in the criminal justice system.

WHAT HAPPENED NEXT?

- ProPublica investigation (2016) exposed the bias.
- The case gained national attention.
- COMPAS remains in use in many jurisdictions, but with increased scrutiny and calls for reform.

KEY TAKEAWAY AI systems are only as fair as the data they are trained on. When historical injustice is embedded in data, AI can automate and amplify that injustice at scale.

COMPAS Findings

Key Takeaways

- AI can inherit bias from historical data.
- High accuracy does not guarantee fairness.
- Different fairness metrics can lead to different conclusions.
- AI decisions can have significant real-world consequences.
- Transparency and independent auditing are essential for high-stakes AI systems.
- Human oversight remains critical in criminal justice applications.

Discussion Question

If a model is statistically accurate overall but makes more mistakes for one group than another, should it still be considered fair?

Summary

The COMPAS case became one of the most famous examples of AI fairness because it showed that a model can appear accurate overall while still producing unequal outcomes across different groups.

Why COMPAS Matters

	Equal Opportunity	False Positive Rate Equality	Calibration
Question Asked	Are qualified individuals treated equally?	Are mistakes distributed equally?	Do risk scores mean the same thing for everyone?
What Researchers Measured	True Positive Rates	False Positive Rates	Accuracy of risk scores
COMPAS Finding	Different performance across groups	Different error rates across groups	Similar meaning of risk scores across groups
Verdict	❌ Unfair	❌ Unfair	✅ Fair

Should AI Be Used in Criminal Justice?

Examples:

- Bail decisions
- Sentencing recommendations
- Recidivism prediction
- Parole decisions

Yes, AI Can Improve Consistency

- Human judges can be influenced by fatigue, emotions, and unconscious biases.
- AI can apply the same criteria consistently across thousands of cases.
- When properly audited and monitored, AI may reduce some forms of human bias.

Key Idea:

AI can support more consistent decision-making than humans alone.

No, The Stakes Are Too High

- Criminal justice decisions directly affect freedom, livelihoods, and human rights.
- Errors can have severe consequences for individuals and families.
- If an AI system is biased or makes mistakes, those mistakes may be scaled across entire populations.

Key Idea:

High-impact decisions require a higher standard of fairness and accountability.

AI Should Assist, Not Replace Humans

- AI can identify patterns and provide risk assessments.
- Human judges should make the final decision and remain accountable.
- AI should be treated as a decision-support tool rather than an autonomous decision-maker.

Key Idea:

Human-in-the-loop approaches often provide the best balance between efficiency and accountability.

Mitigating Bias

Approach	When Applied	Simple Idea
Pre-processing	Before training	Improve the dataset before the model learns from it.
In-processing	During training	Modify the learning algorithm to consider fairness.
Post-processing	After training	Adjust the model's outputs to improve fairness.

Reweighing: Increase Influence of Underrepresented Groups

Give more weight to underrepresented groups so the model learns from a more balanced view.

1. Original Training Data

Underrepresented group has less influence.

Majority Group
80% of data

Underrepresented Group
20% of data

2. Reweighing

Increase the weight of underrepresented groups.

Assign higher weights to examples from the underrepresented group.

3. Reweighted Training Data

Underrepresented group has more influence.

Majority Group
80% of data

Underrepresented Group
20% of data

What This Achieves

- Model pays more attention to underrepresented examples.
- Reduces bias caused by imbalanced training data.
- Leads to fairer predictions for all groups.
- Improves fairness without discarding valuable data.

Key Idea: Reweighing doesn't change the data— it changes the importance (weight) of each example so underrepresented groups have a stronger voice in model learning.

Resampling: Amazon Hiring Example
Balance the training data before building the model.
Amazon's hiring AI was trained on ~10 years of historical hiring data.
Most successful applicants were men, so the model saw more male examples.
1 BEFORE RESAMPLING (Imbalanced Data)
Men Successful Resumes 800
Women Successful Resumes 200
Problem: The model sees 4x more successful male resumes → Male patterns dominate learning.
2 AFTER OVERSAMPLING (Balanced Data)
Men Training Examples 800
Women Training Examples 800
What changed? Female examples are duplicated (or synthetically generated) so both groups contribute equally during training.
3 ALTERNATIVE: UNDERSAMPLING
Before (Imbalanced)
Men 800
Women 200
Remove some male examples
After Undersampling
Men 200
Women 200
Tradeoff: Reduces bias, but discards useful data.
WHY RESAMPLING HELPS
More balanced representation of all groups
Reduces historical hiring imbalance
Improves fairness across groups
Simple and easy to implement
LIMITATIONS
Oversampling can lead to overfitting
Undersampling may lose information
Does not completely eliminate bias
KEY TAKEAWAY: Resampling changes the dataset itself. By balancing the number of examples from different groups, the model learns from a more representative view of the population.

Fairness-Aware Training: A potential solution

Training Data
↓
Model Training
↓
Maximize Accuracy
↓
Predictions

→

Training Data
▼
Fairness-Aware Training
Optimize:
- Accuracy
- Fairness
▼
Machine Learning Model
▼
Predictions
▼
Fairness Evaluation

Human Oversight

- AI provides recommendations, not final responsibility.
- Humans review high-impact decisions before action is taken.
- Human oversight helps detect errors, bias, and unusual cases.
- Accountability remains with people and organizations, not the AI system.
- Human-in-the-loop improves trust, fairness, and safety.

Data
|
▼
AI Model
|
Recommendation
|
▼
Human Review
|
Approve / Modify / Reject
|
▼
Final Decision